

X-BAND DIGITAL MODULE

20x20cm Space-Grade

Direct RF Sampling Acquisition System

Complete Design Package

Revision 1.0 | 2026-09-10

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KEY SPECIFICATIONS

Architecture	Direct RF Sampling
ADC	ADC12DJ5200-SP (10.4 GSPS, 12-bit, rad-hard)
FPGA	XQRVC1902 (rad-hard, 1760-pin BGA)
RF Bandwidth	500 MHz (8-12 GHz X-band)
DDC Channels	8 independent channels
Data Output	SpaceWire (2 links, 100 Mbps each)
Power Input	4.5V DC (space bus)
Power Rails	6 rails (1.0V, 1.8V, 3.3V, 1.0V, 1.8V, 2.5V)
Total Power	61.8W input, 50.3W output (81.4% efficiency)
Board Size	200mm x 200mm x 28mm
Layers	12-layer Megtron6 stackup
Temperature	-55C to +125C (MIL-STD-883)
Radiation	100 krad(Si) TID, SEL immune
MTBF	259,000 hours (29.6 years)
Mass	< 1.2 kg (estimated)

POWER BUDGET SUMMARY

Rail	Vout	I _{typ}	I _{max}	P _{typ}	P _{max}	Type
VCCINT	1.0V	30A	44A	30W	44W	Buck
VCCAUX	1.8V	4A	6A	7.2W	10.8W	Buck
VCCO	3.3V	2A	3A	6.6W	9.9W	Buck
AVDD	1.0V	1A	1.5A	1W	1.5W	LDO
DVDD	1.8V	1A	1.5A	1.8W	2.7W	LDO
VCLK	2.5V	1.5A	2A	3.75W	5W	LDO

COMPONENT SUMMARY

Category	Count	Cost (USD)	Notes
ICs (Rad-Hard)	14	\$101,620	ADC, FPGA, Clock, Power
Oscillator	1	\$185	Crystek CVHD-950 VCXO
Resistors	50	\$1	0402/0603, 1%/0.1%
Capacitors	168	\$49	C0G/X7R/X5R/Tantalum
Inductors	3	\$17.50	100nH/2.2uH/4.7uH
Ferrite Beads	11	\$0.55	Power filtering
Protection	6	\$1.40	TVS, ESD diodes
Connectors	9	\$109.50	SMA, SpaceWire, JTAG
Mechanical	10	\$161	Standoffs, shield, HS
TOTAL	272	\$102,144	